INTRO TO VLSI

ASSIGNMENT-3



AVYAAZ - EE23BTECH11013

AUGUST 2026

Problem 1

Make schematic and layout of a static CMOS Inverter with proper DRC + LVS + Antenna Error clean. Run the RC extraction of the same and compare with schematic result. Find the maximum switching frequency for both cases.

Solution

From the Fig. 1, schematic shows a static CMOS inverter, consisting of a PMOS connected to VDD and an NMOS connected to GND. The input is applied to both gates, and the output is taken from the common drain node.

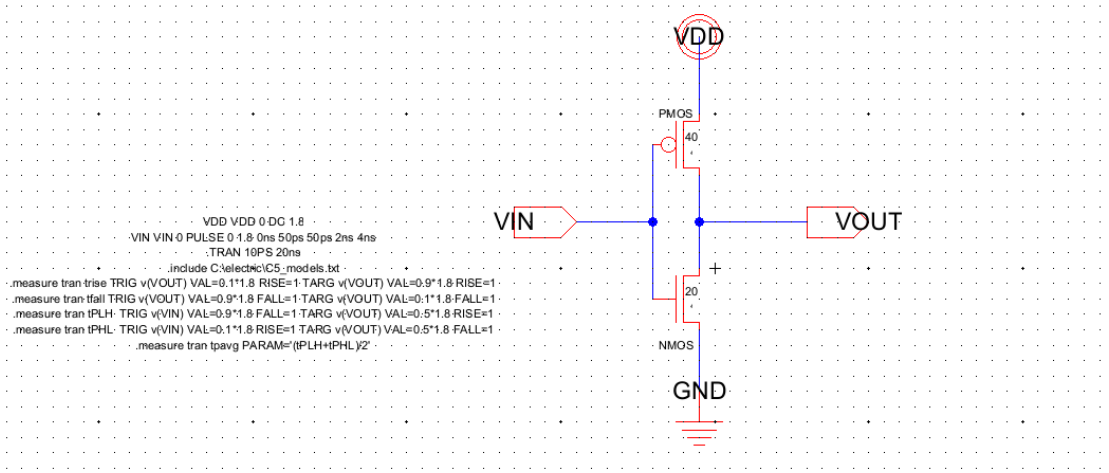


Figure 1: *Schematic of CMOS Inverter*

From Fig. 2, we can observe some undershoots and overshoots at the transitions, which can be avoided by adding a capacitor at the load.

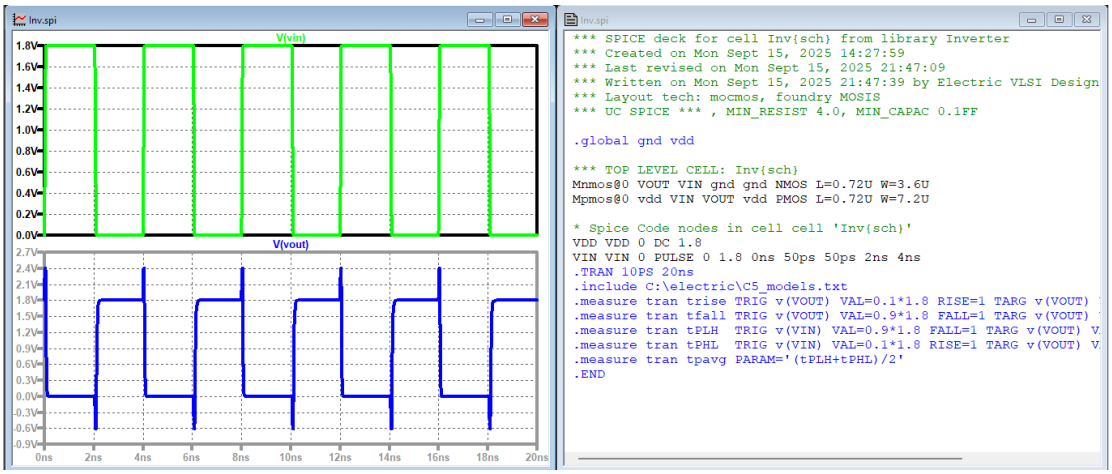


Figure 2: *Input and Output waveforms from layout of cmos inverter*

Below(Fig. 3) are the t_{rise}, t_{fall}, $t_{p,avg}$ measurements from the LTSpice:

```
trise=5.05312e-11 FROM 2.11471e-09 TO 2.16524e-09
tfall=3.35964e-11 FROM 5.82028e-11 TO 9.17992e-11
tplh=7.53585e-11 FROM 2.055e-09 TO 2.13036e-09
tphl=6.35856e-11 FROM 5e-12 TO 6.85856e-11
tpavg: ((tplh+tphl)/2)=6.94721e-11
```

Figure 3: *Delay measurements from the schematic simulation*

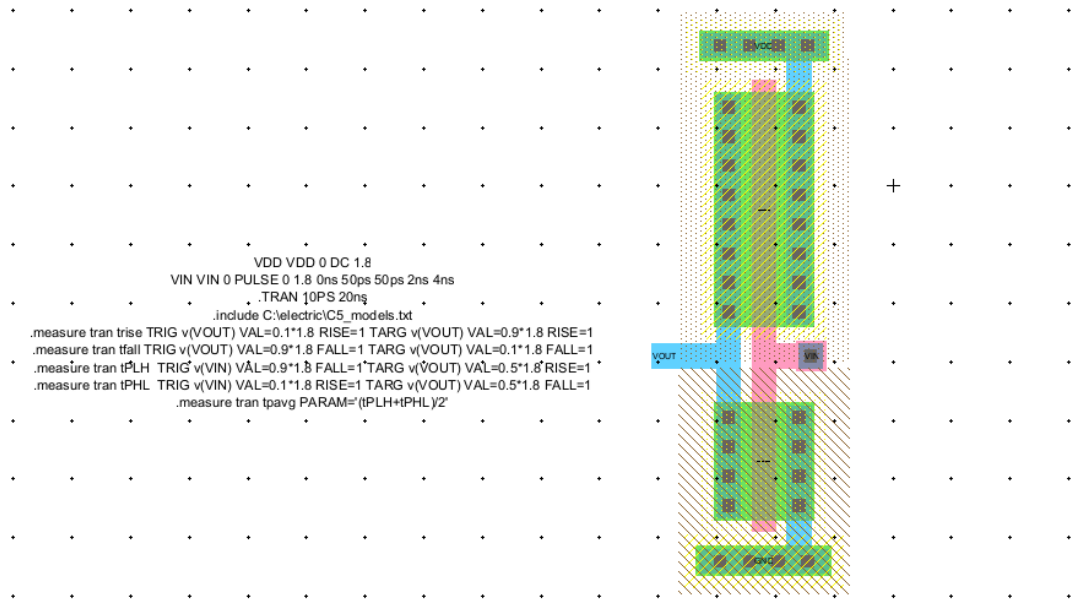


Figure 4: *Layout of CMOS Inverter*

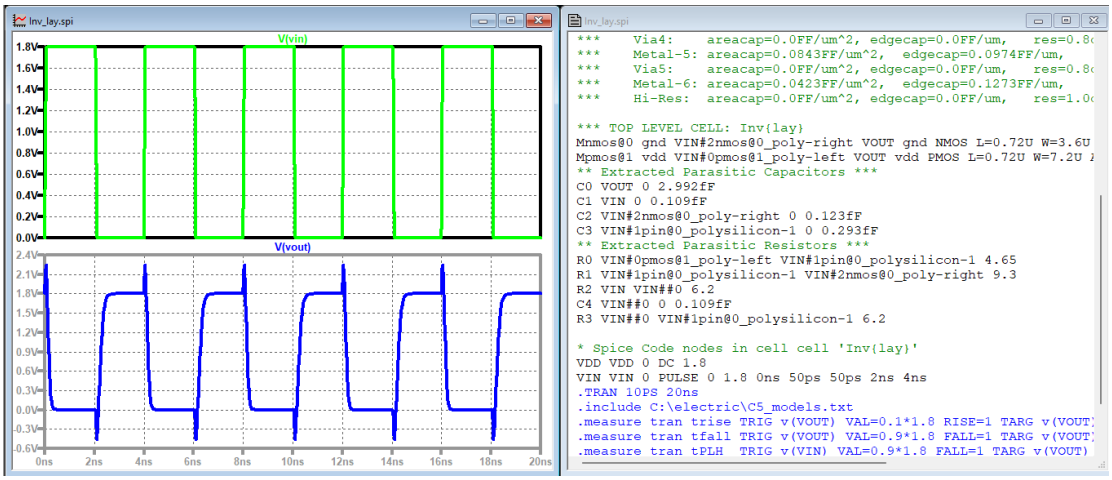


Figure 5: *Input and Output waveforms from schematic of cmos inverter with RC Extraction*

The layout of the CMOS inverter is designed ensuring proper spacing, width, and enclosure rules to pass Design Rule Checking (DRC). The layout is then verified using Layout vs. Schematic (LVS) to ensure it matches the schematic. Antenna errors are also mitigated.

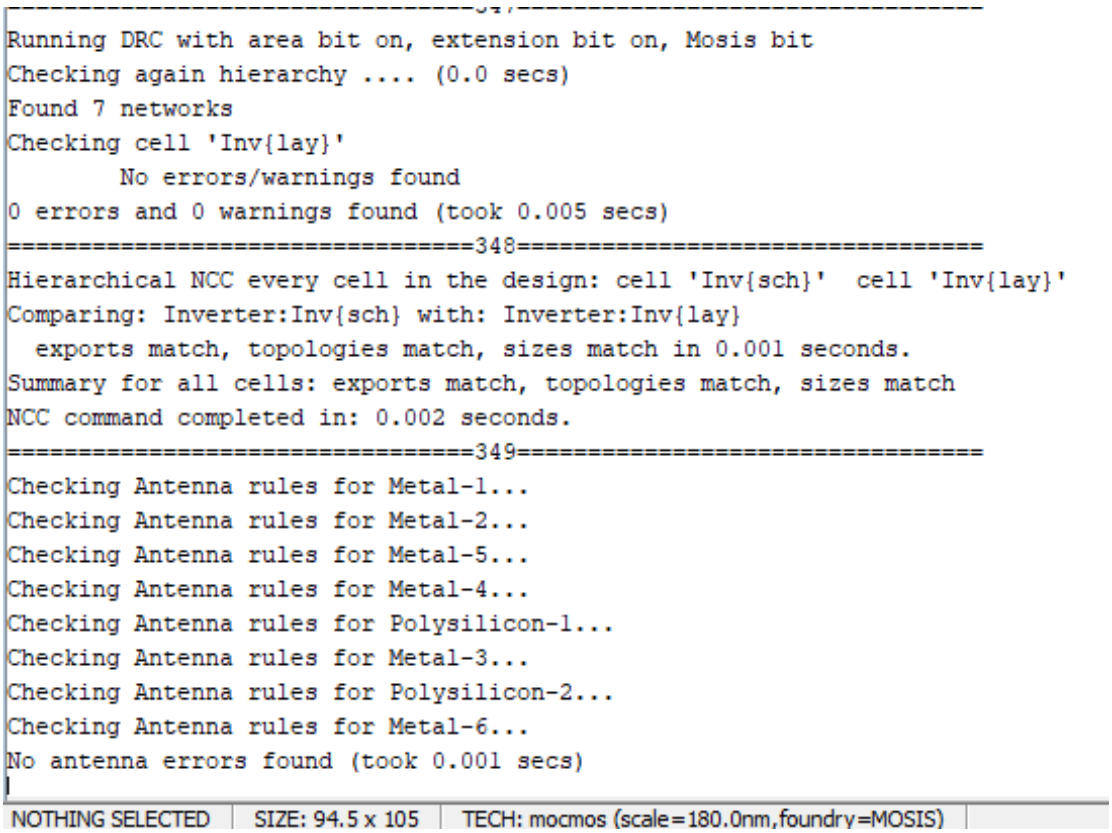


Figure 6: *Layout with DRC + LVS + Antenna Error clean*

The propagation delays(t_{phl} and t_{plh}) are measured for both schematic and layout simulations. The RC-extracted layout shows increased delay due to parasitic capacitances and resistances.

```
trise=2.29555e-10 FROM 2.18001e-09 TO 2.40957e-09
tfall=1.49689e-10 FROM 1.03629e-10 TO 2.53318e-10
tplh=2.13356e-10 FROM 2.055e-09 TO 2.26836e-09
tphl=1.58395e-10 FROM 5e-12 TO 1.63395e-10
tpavg: ((tplh+tphl)/2)=1.85875e-10
```

Figure 7: *Delay measurements from the schematic simulation with RC Extraction*

Including the RC Extraction,

- Maximum switching frequency in the schematic is $\frac{1}{6.94 \times 10^{-11}} = 14.4 \times 10^9$Hz.
- Maximum switching frequency in the layout is $\frac{1}{1.8587 \times 10^{-10}} = 5.38 \times 10^9$Hz.

Problem 2

Repeat Q1 for 3 input NAND and NOR gate with proper sizing.

Solution

1. NAND GATE

The CMOS NAND gate consists of two PMOS transistors in parallel and two NMOS transistors in series. The Boolean expression is given by:

The pull-up network (PUN) comprises PMOS transistors, and the pull-down network (PDN) comprises NMOS transistors.

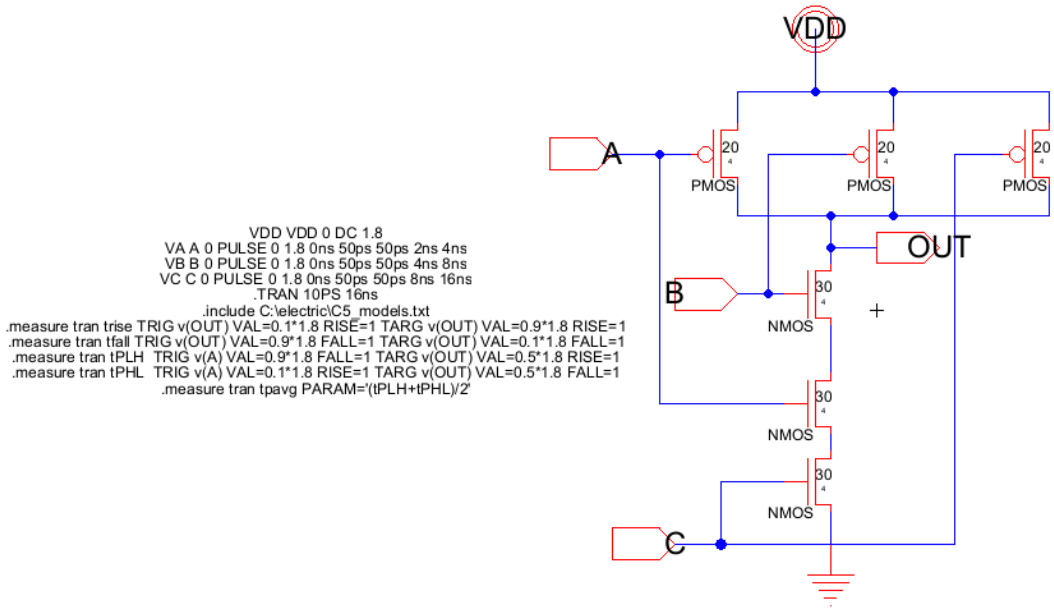


Figure 8: *Schematic of CMOS NAND GATE*

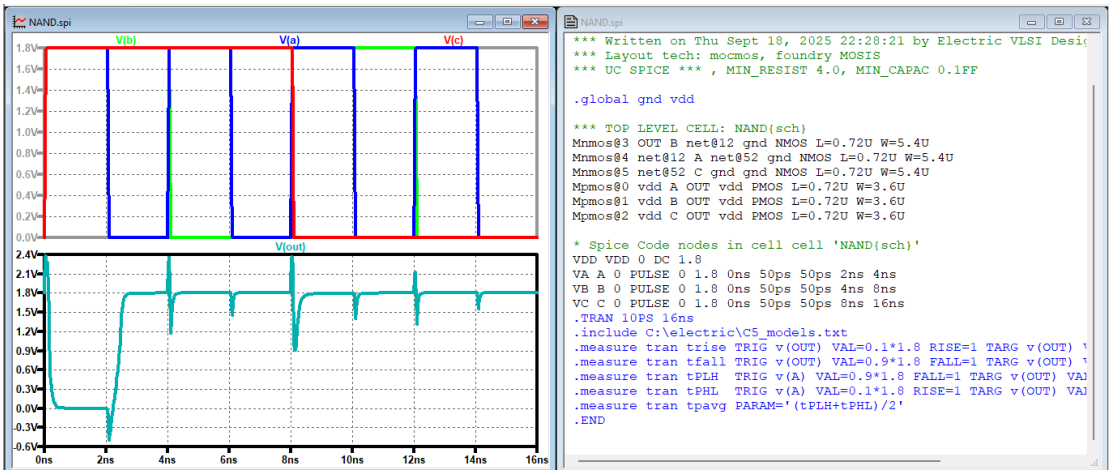


Figure 9: *Input and Output waveforms from layout of cmos NAND GATE*

```
trise=2.67712e-10 FROM 2.2696e-09 TO 2.53731e-09
tfall=1.34059e-10 FROM 1.32918e-10 TO 2.66977e-10
tplh=3.79991e-10 FROM 2.055e-09 TO 2.43499e-09
tphl=1.5643e-10 FROM 5e-12 TO 1.6143e-10
tpavg: ((tplh+tphl)/2)=2.68211e-10
```

Figure 10: *Delay measurements from the schematic simulation*

```
VDD VDD 0 DC 1.8
VA A 0 PULSE 0 1.8 0ns 50ps 50ps 2ns 4ns
VB B 0 PULSE 0 1.8 0ns 50ps 50ps 4ns 8ns
VC C 0 PULSE 0 1.8 0ns 50ps 50ps 8ns 16ns
.TRAN 10PS 10ns
.include C:\electric\C5_models.txt
.measure tran trise TRIG v(OUT) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.9*1.8 RISE=1
.measure tran tfall TRIG v(OUT) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.1*1.8 FALL=1
.measure tran tPLH  TRIG v(A) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.5*1.8 RISE=1
.measure tran tPHL  TRIG v(A) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.5*1.8 FALL=1
.measure tran tpavg PARAM='(tPLH+tPHL)/2'
```

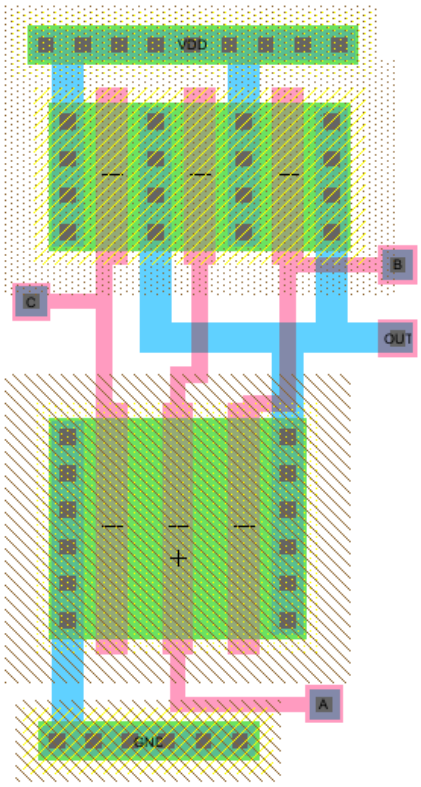


Figure 11: *Layout of CMOS NAND GATE*

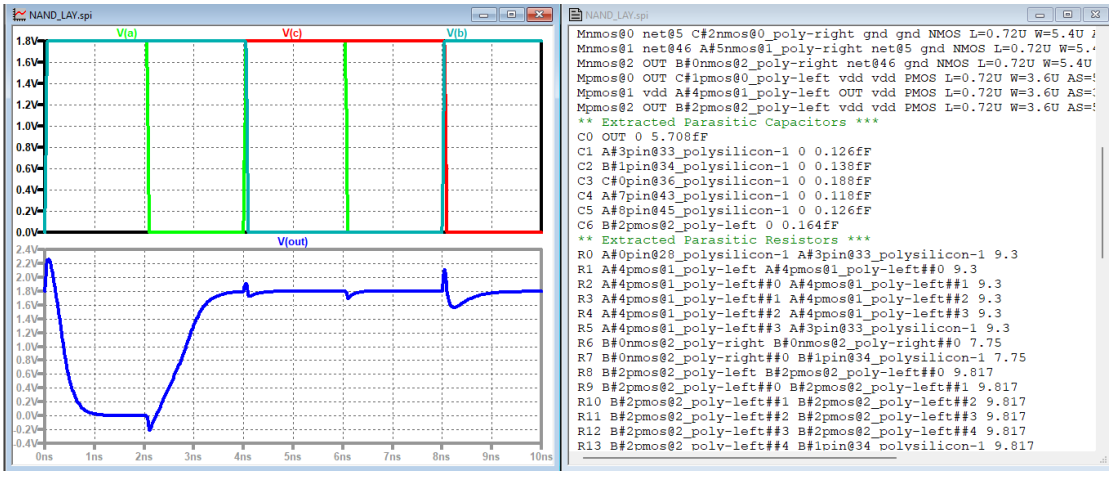


Figure 12: *Input and Output waveforms from schematic of cmos NAND GATE with RC EXTRACTION*

```
Running DRC with area bit on, extension bit on, Mosis bit
Checking again hierarchy .... (0.0 secs)
Found 11 networks
0 errors and 0 warnings found (took 0.002 secs)
================================275================================
Hierarchical NCC every cell in the design: cell 'NAND{sch}'  cell 'NAND{lay}'
Comparing: NANDGATE:NAND{sch} with: NANDGATE:NAND{lay}
  exports match, topologies match, sizes match in 0.003 seconds.
Summary for all cells: exports match, topologies match, sizes match
NCC command completed in: 0.004 seconds.
================================276================================
Checking Antenna rules for Metal-1...
Checking Antenna rules for Metal-2...
Checking Antenna rules for Metal-5...
Checking Antenna rules for Metal-4...
Checking Antenna rules for Polysilicon-1...
Checking Antenna rules for Metal-3...
Checking Antenna rules for Polysilicon-2...
Checking Antenna rules for Metal-6...
No antenna errors found (took 0.004 secs)
```

Figure 13: *Layout with DRC + LVS + Antenna Error clean*

```
trise=8.9697e-10 FROM 2.3493e-09 TO 3.24627e-09
tfall=4.29515e-10 FROM 2.50306e-10 TO 6.79821e-10
tplh=7.35502e-10 FROM 2.055e-09 TO 2.7905e-09
tphl=3.92911e-10 FROM 5e-12 TO 3.97911e-10
tpavg: ((tplh+tphl)/2)=5.64207e-10
```

Figure 14: *Delay measurements from the schematic simulation with RC EXTRACTION*

Including the RC Extraction,

- Maximum switching frequency in the schematic is $\frac{1}{2.68 \times 10^{-10}} = 3.73 \times 10^{9}$Hz.
- Maximum switching frequency in the layout is $\frac{1}{5.642 \times 10^{-10}} = 1.77 \times 10^{9}$Hz.

2.NOR GATE

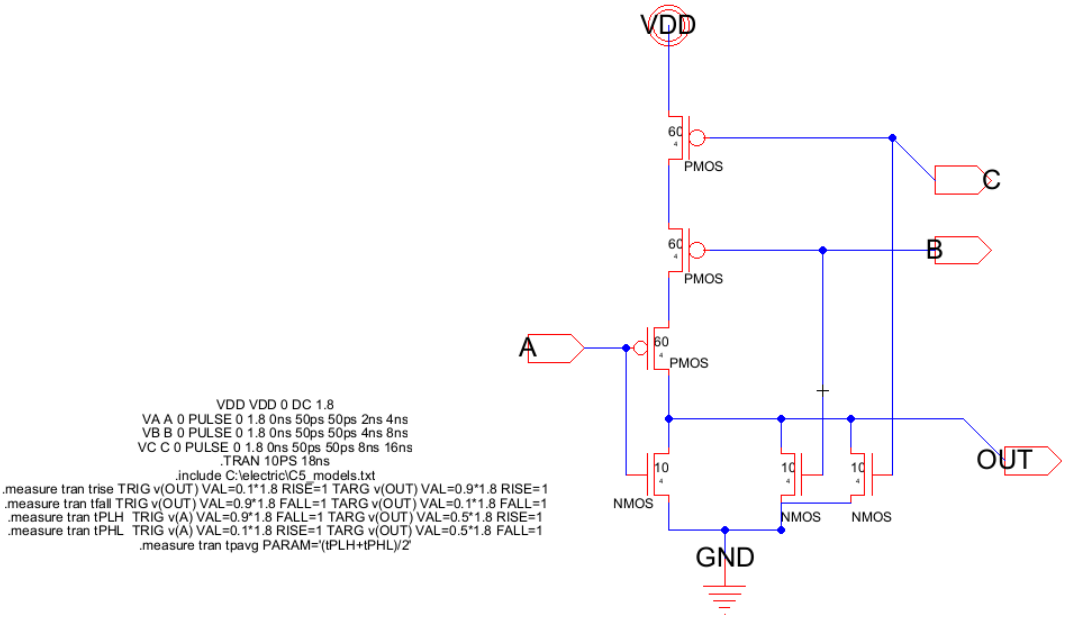


Figure 15: *Schematic of CMOS NOR GATE*

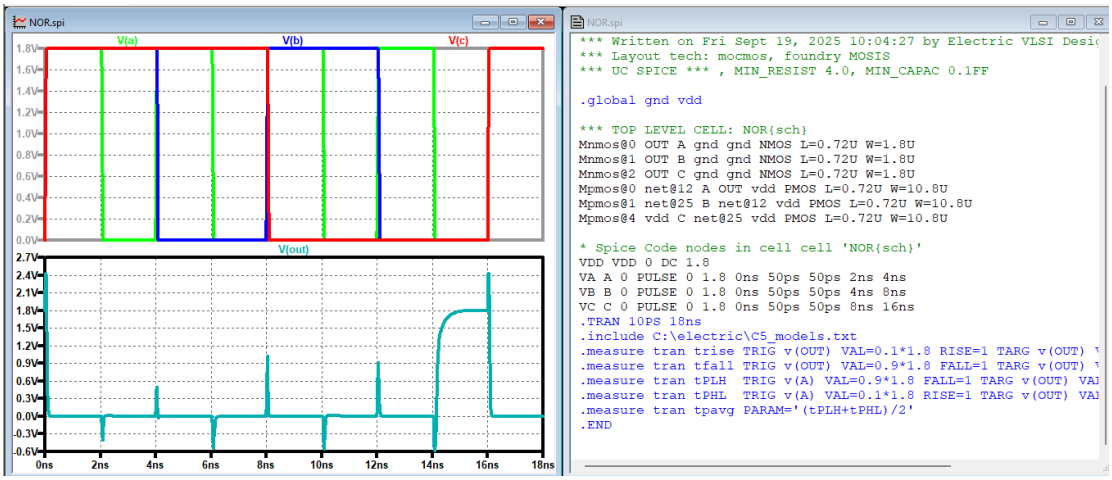


Figure 16: *Input and Output waveforms from layout of cmos NOR GATE*

```
trise=1.04347e-08 FROM 4.01149e-09 TO 1.44462e-08
tfall=4.62495e-11 FROM 6.43117e-11 TO 1.10561e-10
tplh=5.9827e-09 FROM 2.055e-09 TO 8.0377e-09
tphl=7.52942e-11 FROM 5e-12 TO 8.02942e-11
tpavg: ((tplh+tphl)/2)=3.029e-09
```

Figure 17: *Delay measurements from the schematic simulation*

```
VDD VDD 0 DC 1.8
VA A 0 PULSE 0 1.8 0ns 50ps 50ps 2ns 4ns
VB B 0 PULSE 0 1.8 0ns 50ps 50ps 4ns 8ns
VC C 0 PULSE 0 1.8 0ns 50ps 50ps 8ns 16ns
.TRAN 10PS 10ns
.include C:\electric\C5_models.txt
.measure tran trise TRIG v(OUT) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.9*1.8 RISE=1
.measure tran tfall TRIG v(OUT) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.1*1.8 FALL=1
.measure tran tPLH  TRIG v(A) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.5*1.8 RISE=1
.measure tran tPHL  TRIG v(A) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.5*1.8 FALL=1
.measure tran tpavg PARAM='(tPLH+tPHL)/2'
```

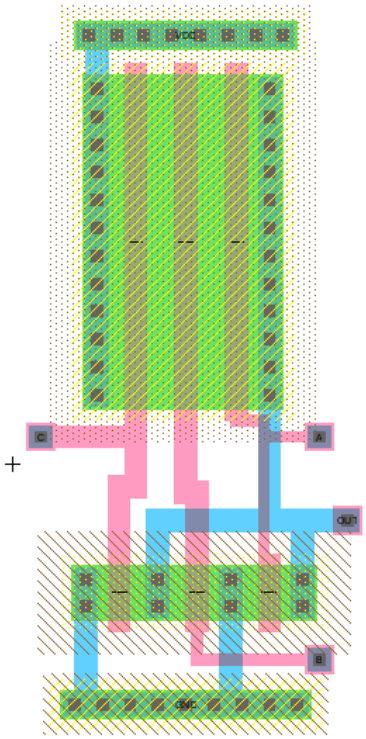


Figure 18: *Layout of CMOS NOR GATE*

```
Running DRC with area bit on, extension bit on, Mosis bit
Checking again hierarchy .... (0.0 secs)
Found 15 networks
Checking cell 'NOR{lay}'
        No errors/warnings found
0 errors and 0 warnings found (took 0.093 secs)
=================================9=================================
Hierarchical NCC every cell in the design: cell 'NOR{sch}'  cell 'NOR{lay}'
Comparing: NORGATE:NOR{sch} with: NORGATE:NOR{lay}
  exports match, topologies match, sizes match in 0.048 seconds.
Summary for all cells: exports match, topologies match, sizes match
NCC command completed in: 0.058 seconds.
=================================10================================
Checking Antenna rules for Metal-5...
Checking Antenna rules for Metal-3...
Checking Antenna rules for Metal-6...
Checking Antenna rules for Polysilicon-2...
Checking Antenna rules for Metal-1...
Checking Antenna rules for Polysilicon-1...
Checking Antenna rules for Metal-2...
Checking Antenna rules for Metal-4...
No antenna errors found (took 0.01 secs)
```

Figure 19: *Layout with DRC + LVS + Antenna Error clean*

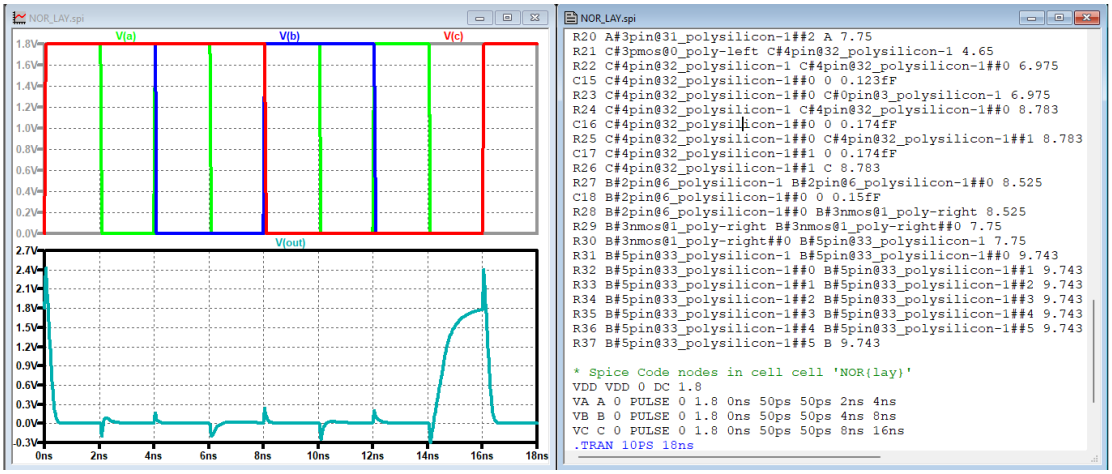


Figure 20: *Input and Output waveforms from schematic of cmos NOR GATE with RC EXTRACTION*

```
trise=1.11199e-08 FROM 4.04932e-09 TO 1.51692e-08
tfall=2.26074e-10 FROM 1.51879e-10 TO 3.77952e-10
tplh=1.25081e-08 FROM 2.055e-09 TO 1.45631e-08
tphl=2.39095e-10 FROM 5e-12 TO 2.44095e-10
tpavg: ((tplh+tphl)/2)=6.37362e-09
```

Figure 21: *Delay measurements from the schematic simulation with RC EXTRACTION*

Including the RC Extraction,

- Maximum switching frequency in the schematic is $\frac{1}{3.029 \times 10^{-9}} = 0.33 \times 10^{9}$Hz.
- Maximum switching frequency in the layout is $\frac{1}{6.373 \times 10^{-10}} = 0.157 \times 10^{9}$Hz.

Problem 3

Repeat Q1 and Q2 for a load capacitance of 1 pF.

Solution

Note: I had to change the input pulses duration as the time constant required to charge and discharge for the circuit is very high but provided load time constant is very low. So, the output waveform isn't able to reach the low level or high level completely.

1.Inverter

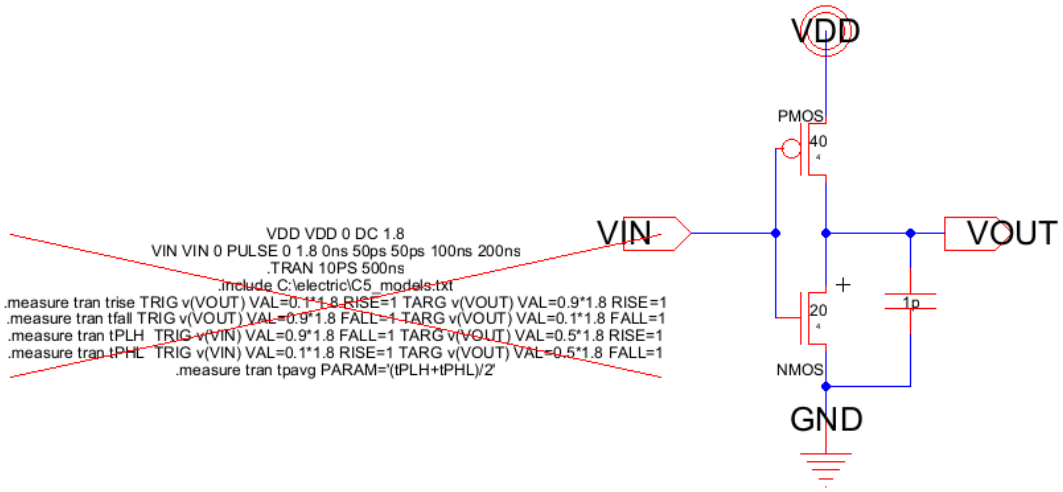


Figure 22: *Schematic of CMOS Inverter*

From Fig. 23, we can observe some undershoots and overshoots at the transitions before are removed by adding a capacitor at the load.

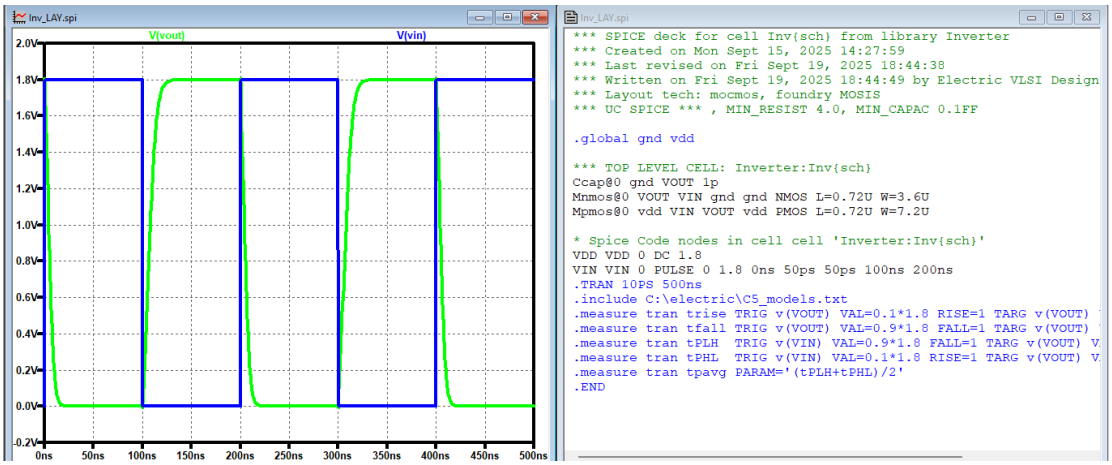


Figure 23: *Input and Output waveforms from layout of cmos inverter*

Below(Fig. 24) are the t_{rise}, t_{fall}, $t_{p,avg}$ measurements from the LTSpice:

```
trise=1.38198e-08 FROM 1.01519e-07 TO 1.15339e-07
tfall=9.1387e-09 FROM 1.00206e-09 TO 1.01408e-08
tplh=7.14503e-09 FROM 1.00055e-07 TO 1.072e-07
tphl=4.77513e-09 FROM 5e-12 TO 4.78013e-09
tpavg: ((tplh+tphl)/2)=5.96008e-09
```

Figure 24: *Delay measurements from the schematic simulation*

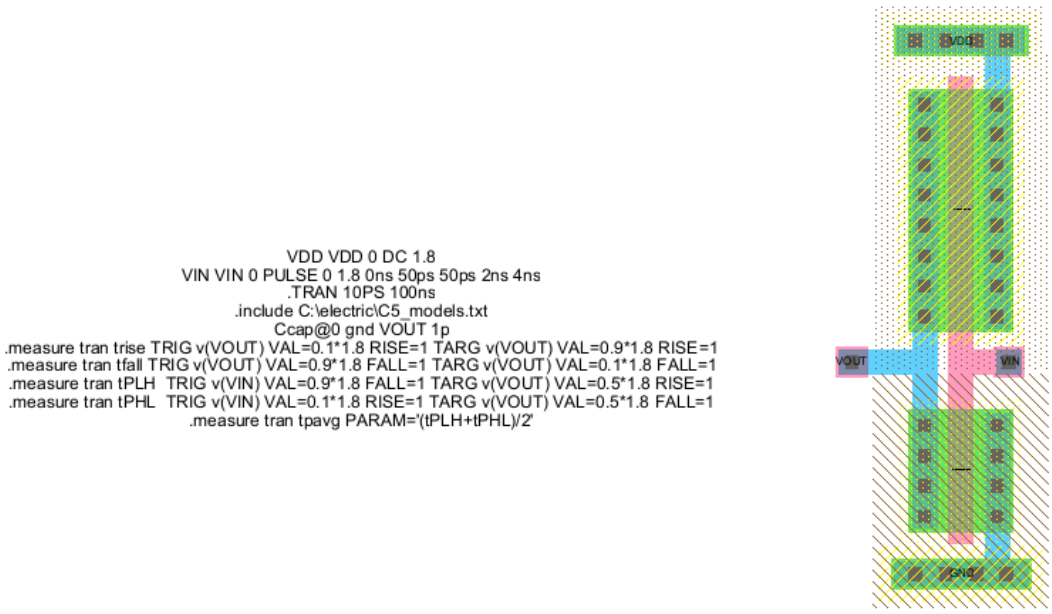


Figure 25: *Layout of CMOS Inverter*

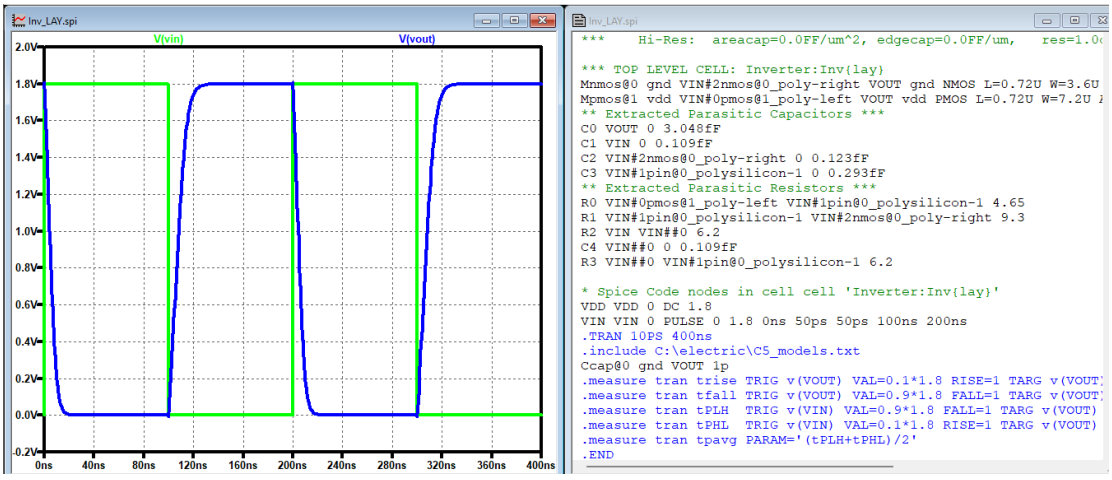


Figure 26: *Input and Output waveforms from schematic of cmos inverter with RC Extraction*

The propagation delays(t_{phl} and t_{plh}) are measured for both schematic and layout

simulations. The RC-extracted layout shows increased delay due to parasitic capacitances and resistances.

```
trise=1.40055e-08 FROM 1.01536e-07 TO 1.15542e-07
tfall=9.24157e-09 FROM 1.01888e-09 TO 1.02604e-08
tplh=7.23766e-09 FROM 1.00055e-07 TO 1.07293e-07
tphl=4.83727e-09 FROM 5e-12 TO 4.84227e-09
tpavg: ((tplh+tphl)/2)=6.03747e-09
```

Figure 27: *Delay measurements from the schematic simulation with RC Extraction*

Including the RC Extraction,

- Maximum switching frequency in the schematic is $\frac{1}{5.96 \times 10^{-9}} = 0.167 \times 10^{9}$Hz.
- Maximum switching frequency in the layout is $\frac{1}{6.037 \times 10^{-9}} = 0.165 \times 10^{9}$Hz.

2.NAND GATE

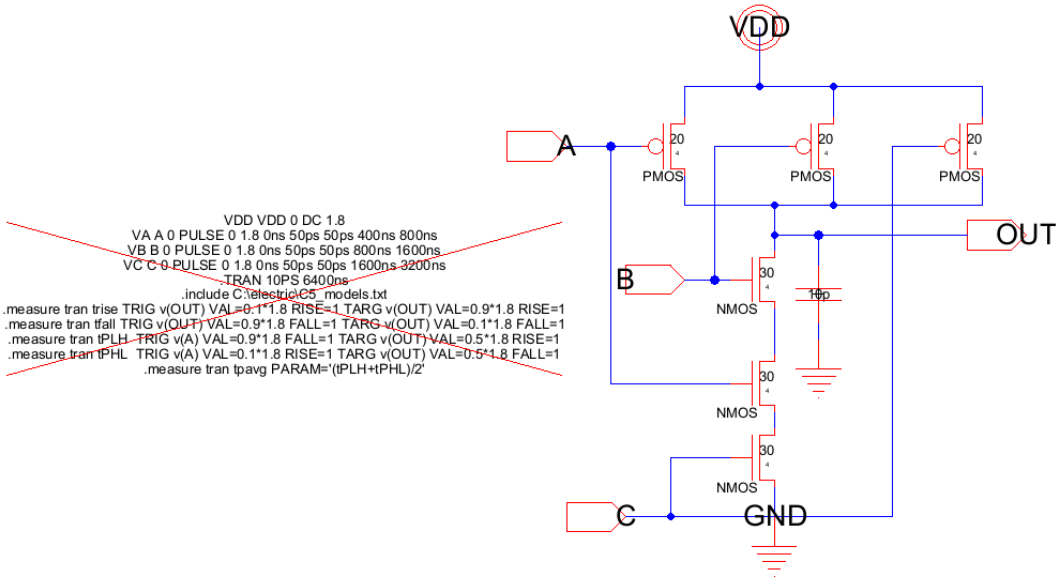


Figure 28: *Schematic of CMOS NAND GATE*

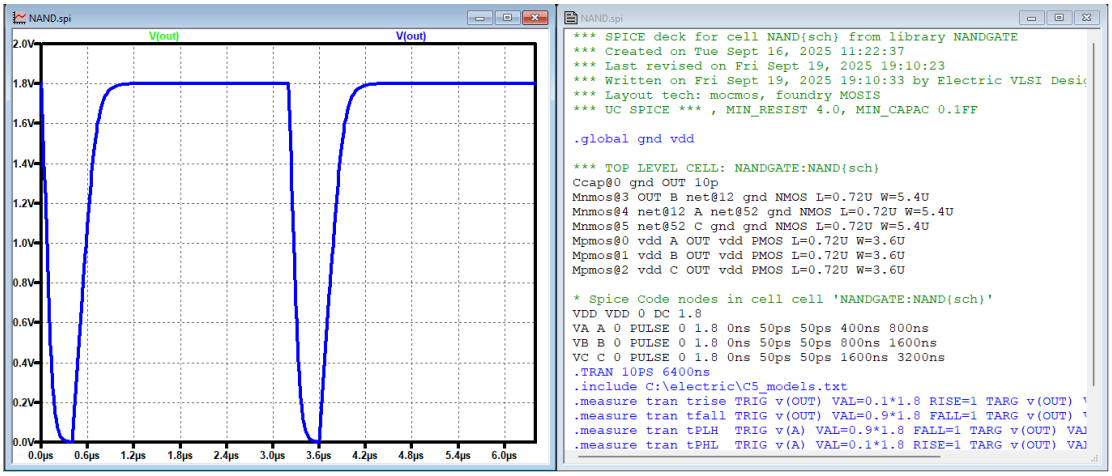


Figure 29: *Input and Output waveforms from layout of cmos NAND GATE*

```
trise=3.07619e-07 FROM 4.30293e-07 TO 7.37913e-07
tfall=1.55326e-07 FROM 1.57165e-08 TO 1.71042e-07
tplh=1.57528e-07 FROM 4.00055e-07 TO 5.57583e-07
tphl=7.90503e-08 FROM 5e-12 TO 7.90553e-08
tpavg: ((tplh+tphl)/2)=1.18289e-07
```

Figure 30: *Delay measurements from the schematic simulation*

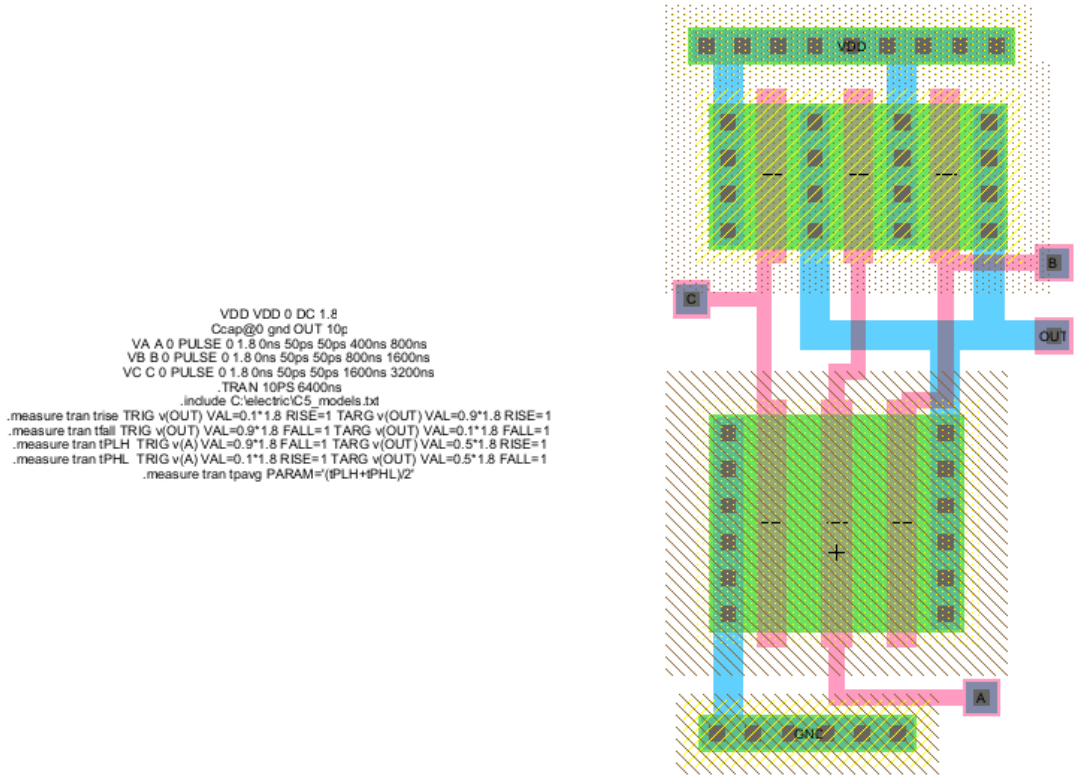


Figure 31: *Layout of CMOS NAND GATE*

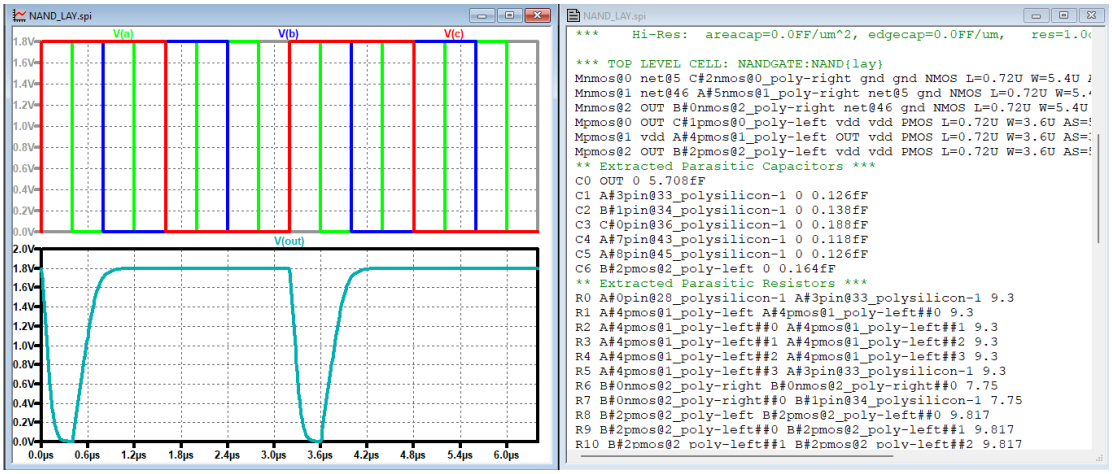


Figure 32: *Input and Output waveforms from schematic of cmos NAND GATE with RC EXTRACTION*

```
trise=3.08423e-07 FROM 4.30356e-07 TO 7.38779e-07
tfall=1.55598e-07 FROM 1.57491e-08 TO 1.71347e-07
tplh=1.57836e-07 FROM 4.00055e-07 TO 5.57891e-07
tphl=7.9199e-08 FROM 5e-12 TO 7.9204e-08
tpavg: ((tplh+tphl)/2)=1.18517e-07
```

Figure 33: *Delay measurements from the schematic simulation with RC EXTRACTION*

Including the RC Extraction,

- Maximum switching frequency in the schematic is $\frac{1}{1.8448 \times 10^{-7}} = 5.42 \times 10^{6}$Hz.
- Maximum switching frequency in the layout is $\frac{1}{1.8517 \times 10^{-7}} = 5.4 \times 10^{6}$Hz.

3.NOR GATE

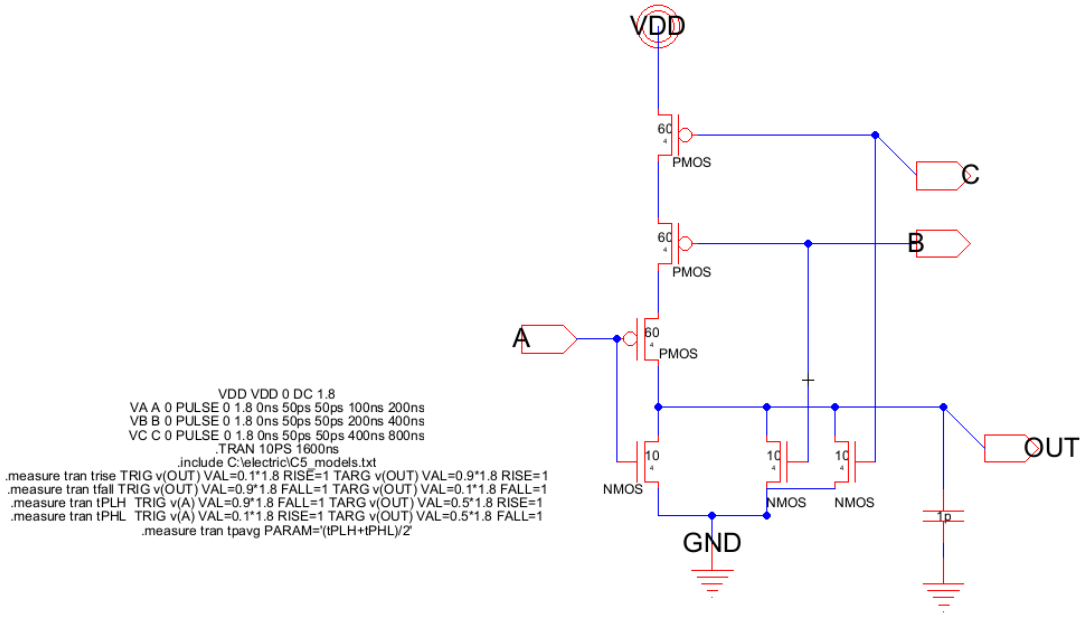


Figure 34: *Schematic of CMOS NOR GATE*

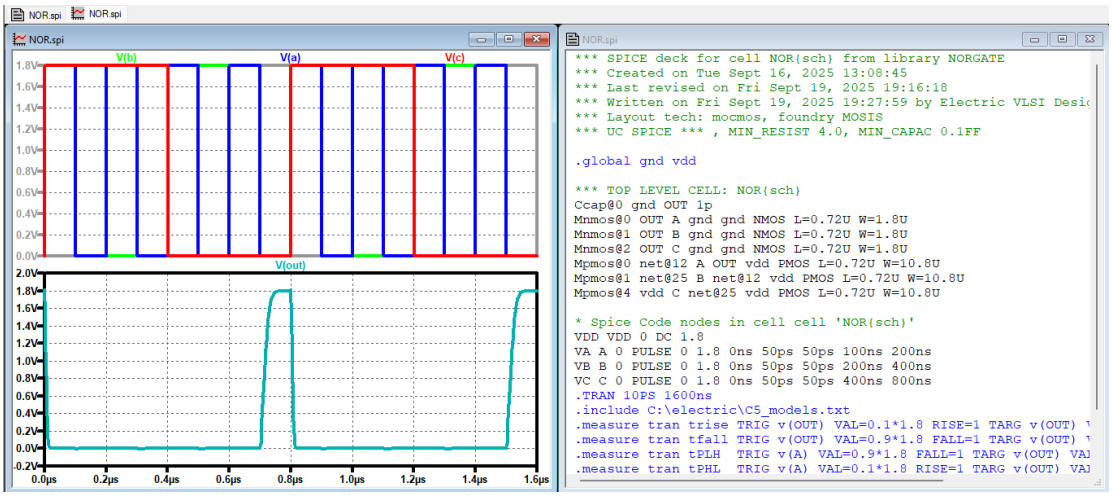


Figure 35: *Input and Output waveforms from layout of cmos NOR GATE*

```
trise=2.81958e-08 FROM 7.03178e-07 TO 7.31374e-07
tfall=9.49913e-09 FROM 1.0973e-09 TO 1.05964e-08
tplh=6.15124e-07 FROM 1.00055e-07 TO 7.15179e-07
tphl=5.06289e-09 FROM 5e-12 TO 5.06789e-09
tpavg: ((tplh+tphl)/2)=3.10093e-07
```

Figure 36: *Delay measurements from the schematic simulation*

```
VDD VDD 0 DC 1.8
VA A 0 PULSE 0 1.8 0ns 50ps 50ps 100ns 200ns
VB B 0 PULSE 0 1.8 0ns 50ps 50ps 200ns 400ns
VC C 0 PULSE 0 1.8 0ns 50ps 50ps 400ns 800ns
.TRAN 10PS 1600ns
Ccap@0 gnd OUT 1p
.include C:\electric\C5_models.txt
.measure tran trise TRIG v(OUT) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.9*1.8 RISE=1
.measure tran tfall TRIG v(OUT) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.1*1.8 FALL=1
.measure tran tPLH  TRIG v(A) VAL=0.9*1.8 FALL=1 TARG v(OUT) VAL=0.5*1.8 RISE=1
.measure tran tPHL  TRIG v(A) VAL=0.1*1.8 RISE=1 TARG v(OUT) VAL=0.5*1.8 FALL=1
.measure tran tpavg PARAM='(tPLH+tPHL)/2'
```

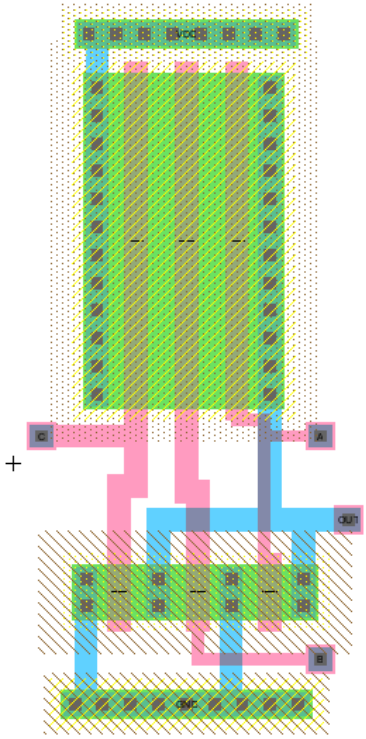


Figure 37: *Layout of CMOS NOR GATE*

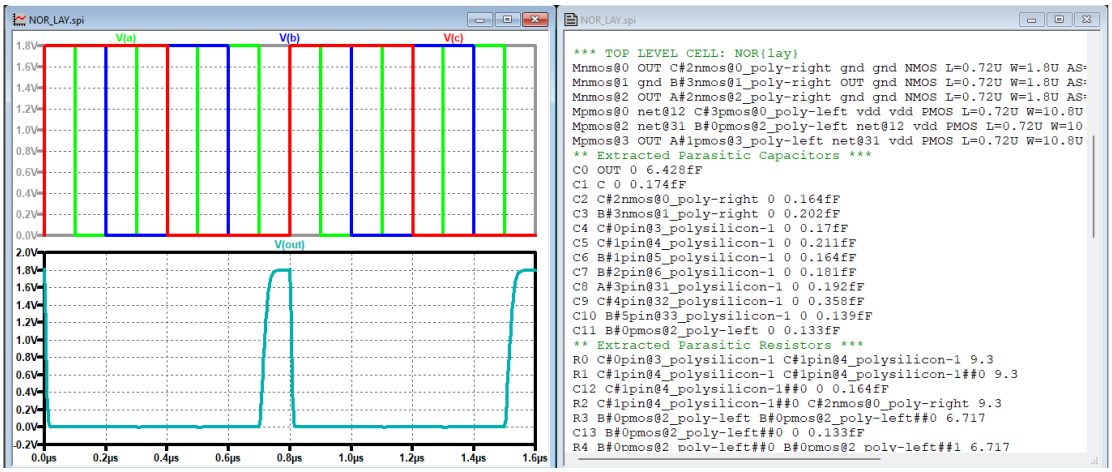


Figure 38: *Input and Output waveforms from schematic of cmos NOR GATE*

```
trise=2.87966e-08 FROM 7.03202e-07 TO 7.31998e-07
tfall=9.63021e-09 FROM 1.11166e-09 TO 1.07419e-08
tplh=6.15393e-07 FROM 1.00055e-07 TO 7.15448e-07
tphl=5.15742e-09 FROM 5e-12 TO 5.16242e-09
tpavg: ((tplh+tphl)/2)=3.10275e-07
```

Figure 39: *Delay measurements from the schematic simulation*

- Maximum switching frequency in the schematic is $\frac{1}{3.1 \times 10^{-7}} = 3.225 \times 10^{6}$Hz.
- Maximum switching frequency in the layout is $\frac{1}{3.10275 \times 10^{-7}} = 3.223 \times 10^{6}$Hz.